DHANYASHREE M V

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PROFESSIONAL SUMMARY

B.E. graduate in Artificial Intelligence & Machine Learning from Dayananda Sagar College of Engineering (CGPA: 9.0/10), combining strong academic performance with hands-on experience in real-world AI/ML projects and research. Adept at breaking down complex problems, applying machine learning and data-driven approaches, designing systematic experiments, and translating research insights into practical solutions. Passionate about building reliable intelligent systems, developing predictive solutions, and exploring emerging advances in AI, with a strong focus on continuous learning and technical innovation.

EDUCATION

B.E. in Artificial Intelligence & Machine Learning *2022 – 2026*
Dayananda Sagar College of Engineering, Bengaluru

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript, HTML, CSS,React.js
ML	Keras,Pytorch,Tensor-Flow,Scikit Learn,Feature Engineering,Model Deployment
GEN AI	LangChain, LangGraph, LLMs, RAG, Scikit-learn, YOLOv11, AI Agents
MLOps	MLflow,Docker,FastAPI, Rest API,CI/CD,Version control(Git)
Databases	PostgreSQL, MongoDB, SQLite,Pinecone
Research & Analysis	Critical Thinking,Data Analysis,Exploratory Data Analysis,Experiment Tracking,LLM Evaluation, AI Evaluation, Prompt Engineering, Ground-Truth Validation, Model Failure Analysis .

PROFESSIONAL EXPERIENCE

Research Intern : IIM Bengaluru CSITM

- Developing an AI-based mobility support device for blind users, integrating real-time object detection to identify obstacles and surroundings
- Building out functionality for education and daily-activity assistance, aiming to support independent use beyond navigation
- Working closely with a professor and interacting with faculty across departments, contributing to the device's system design and testing process

Research & ML Intern - CSIR : National Aerospace Laboratories (NAL)

- Developed a Machine Learning model to predict the Remaining Useful Life (RUL) of aircraft components using real-world strain sensor data for predictive maintenance.
- Performed data preprocessing, feature engineering, and implemented ASTM E1049-85 Rainflow counting with exponential smoothing to extract fatigue features.
- Trained, evaluated, and optimized ML models using Python, validating performance through regression metrics and reproducible experiment workflows.
- Conducted exploratory data analysis on time-series sensor data to identify degradation patterns and improve prediction accuracy.

AI Evaluation Specialist - HANDSHAKE AI - Fellow

- Engineered complex, research-driven evaluation benchmarks to stress-test LLMs on web retrieval, multi-step reasoning, and factual accuracy.
- Performed rigorous multi-source research and ground-truth validation to construct high-confidence datasets for AI model evaluation.
- Diagnosed model hallucinations and reasoning failures through systematic testing, generating targeted evaluation data to improve LLM reliability and search performance.

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PROJECTS

Cortex — Multi-Agent RAG System | LangGraph · LangChain · FastAPI · PostgreSQL · ChromaDB

- Built a production-ready multi-agent AI assistant powered by 9 specialized AI agents orchestrated using LangGraph, combining SQL querying and RAG pipelines to answer academic, administrative, and institutional queries.
- Developed a scalable architecture with a React frontend and FastAPI backend, deployed on Netlify and Railway, enabling real-time interactions and seamless API integration.
- Implemented semantic retrieval using embedding models and vector search, integrated Bulbul v3 Text-to-Speech, Web Speech API for voice interaction, Sarvam AI for multilingual translation, and Sarvam Vision AI for image understanding.
- Optimized retrieval accuracy through prompt engineering, agent orchestration, and hybrid SQL + RAG workflows, delivering context-aware, multilingual, and voice-enabled responses.

WildGuard AI | YOLOv11 · Python · Streamlit · Twilio API · Label Studio

- Developed a YOLOv11-based computer vision system to detect poachers, rangers, tourists, and wildlife from custom camera trap datasets.
- Built an end-to-end ML pipeline including data annotation, preprocessing, model training, evaluation, and experiment tracking using Label Studio and MLflow.
- Developed a Streamlit application for real-time detection with multi-image support and integrated Twilio API for automated SMS alerts on poacher detection.
- Optimized model performance using mAP, precision, and recall, delivering a scalable AI solution for wildlife conservation and surveillance.

Airbnb Smart Pricing – Insights Hub | Python · Scikit-learn · FastAPI · Streamlit · MLflow

- Built an end-to-end machine learning platform to predict Airbnb listing prices using Random Forest, incorporating data preprocessing, feature engineering, and exploratory data analysis on 50K+ listings.
- Developed interactive Streamlit dashboards with Plotly to visualize pricing trends, neighbourhood insights, and key business metrics for data-driven decision-making.
- Deployed the trained model as a FastAPI REST service and tracked experiments, model versions, and performance metrics using MLflow for reproducible ML workflows.
- Optimized model performance through feature selection, hyperparameter tuning, and evaluation using RMSE and R^2, delivering a scalable analytics and prediction platform.

CERTIFICATIONS

- Google Cloud: Advanced - Generative AI for Developers Learning Path & Machine Learning
- Google Cloud Career Launchpad – Cloud Engineer Track(Vertex AI)
- NPTEL – Human Computer Interaction, IIT Madras
- Anthropic Academy — All Courses Completed

LEADERSHIP & ACTIVITIES

- **Content Co-Lead, Aventus 3.0 Hackathon (National-Level) – DSCE Central Committee**
  Led a cross-functional team in planning, content strategy, and event coordination, ensuring timely delivery of communications for 500+ participants.
- **Event Organiser, Innovation & Entrepreneurship Development Cell (IEDC), DSCE**
  Organized technical workshops and innovation events, collaborating with faculty, industry professionals, and student teams to promote entrepreneurship and technology initiatives.